Koç University

College of Engineering

MECH 301 Fluid Mechanics

CFD Project II

Flow and Heat Transfer in a Mixing Elbow

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Abstract

This project presents a computational fluid dynamics (CFD) analysis of a mixing elbow, a pipe structure in power and process industries where effective fluid mixing is essential to prevent thermal fatigue and structural failure. Using ANSYS DesignModeler and Fluent, the study solves the conservation equations of mass, momentum, and energy across both 2D planar and 3D circular domains. The investigation begins with a laminar flow analysis ($Re = 400$) using unstructured meshes, which revealed that mixing is dominated by convection, resulting in distinct thermal stratification and significant flow separation. The study then transitions to turbulent flow using water, employing structured grids where grid independence was rigorously proven through systematic refinement. Furthermore, a comparison between Standard $k - \epsilon$ and SST $k - \omega$ turbulence models yielded consistent velocity profiles, validating the numerical approach. A parametric study varying the small inlet velocity demonstrated a direct positive correlation between jet momentum and mixing quality; higher velocities facilitated deeper jet penetration and enhanced turbulent diffusion. Finally, the 3D analysis captured complex secondary flows, specifically Dean vortices induced by centrifugal forces, which significantly promote lateral mixing not observable in planar simulations. The results confirm that while increasing inlet velocity broadens the temperature distribution, full 3D modeling is indispensable for capturing the realistic vortical structures required for accurate thermal homogeneity predictions.

1. Introduction

This project focuses on the computational fluid dynamics (CFD) analysis of a mixing elbow, which is a common pipe structure used in process plants and power industries where fluids of different temperatures and velocities are mixed. It has 2 inlets and one outlet and throuht this project, it will be clear that the velocity, pressure and temprature contours in the model will behave. Grid convergance will be proved using three different mesh sizes. Different from laminar flow, turbulant flow could also be seen as properties of the fluid changes. 3D results will be compared with 2D result for giving a real word example.

These types of prjects allows engineers to identify potential failure points and optimize system performance. In power plants and process industries, fluids of different temperatures and velocities are frequently mixed. If the temperature distribution is not properly managed, it can lead to thermal fatigue or thermal stress, where repeated temperature fluctuations cause the pipe material to expand and contract, eventually leading to cracks or structural failure. This project will allow

qualitative and quantitative results to be used in real word situations using Ansys DesignModeler and Fluent [3].

2. Description of the Physical Problem

The physical problem analyzed in this project involves the mixing of two fluid streams at different temperatures within a mixing elbow geometry. The domain consists of a large horizontal pipe and a smaller vertical pipe that intersect at a 90-degree elbow, modeled as a 2D planar surface for the initial tasks and a 3D volume with circular cross-sections of 100 mm and 25 mm diameters for the final task Figure 1.

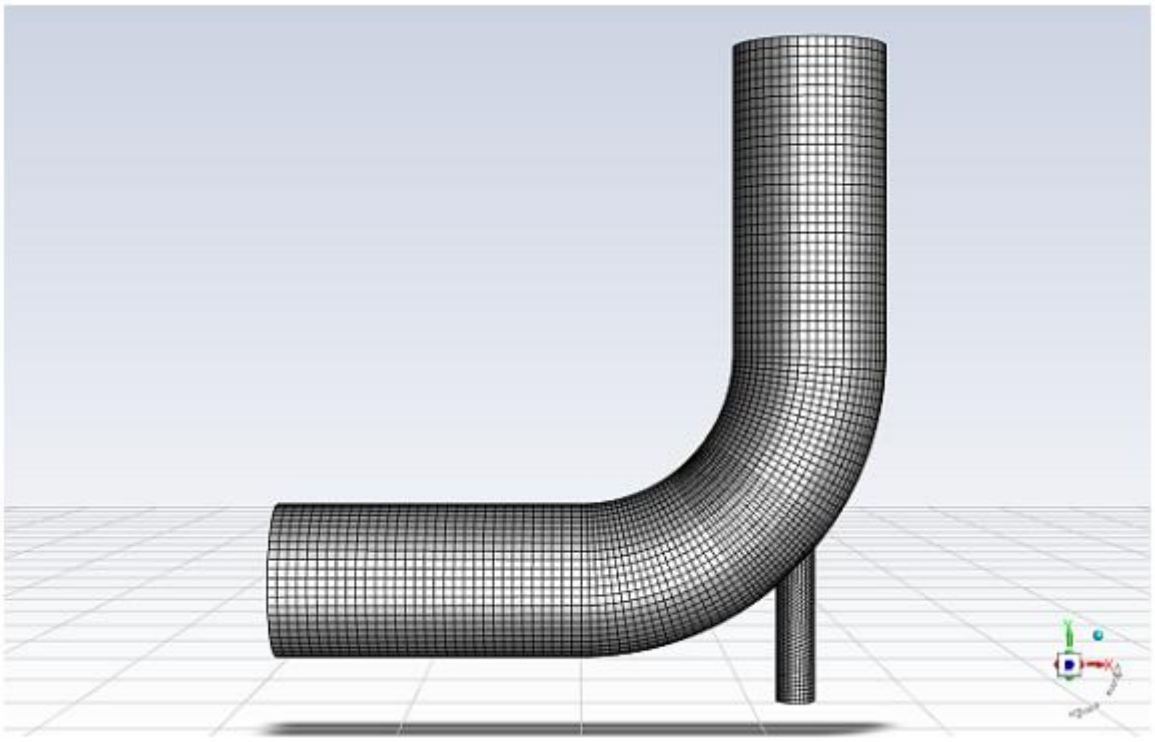


Figure 1. 3D model of the intended pipe from *MECH301_Fall2025_CFDProject2* [2].

Figure 4 is for task 1 and 2 ; Figure 5 is for task 3. For the laminar flow study, the working fluid is defined with a density of 100 kg/m^3, a viscosity of 0.01 $kg/m \times s$, a thermal conductivity of 0.1 $W/m \times K$, and a specific heat capacity of 1.2 $J/kg - K$.

In the subsequent turbulent flow studies, the fluid properties are changed to represent water, characterized by a density of 1000 kg/m^3, a viscosity of 0.001 kg/m-s, a thermal conductivity of 0.67 W/m-K, and a specific heat of 4200 J/kg-K see Figure 2.

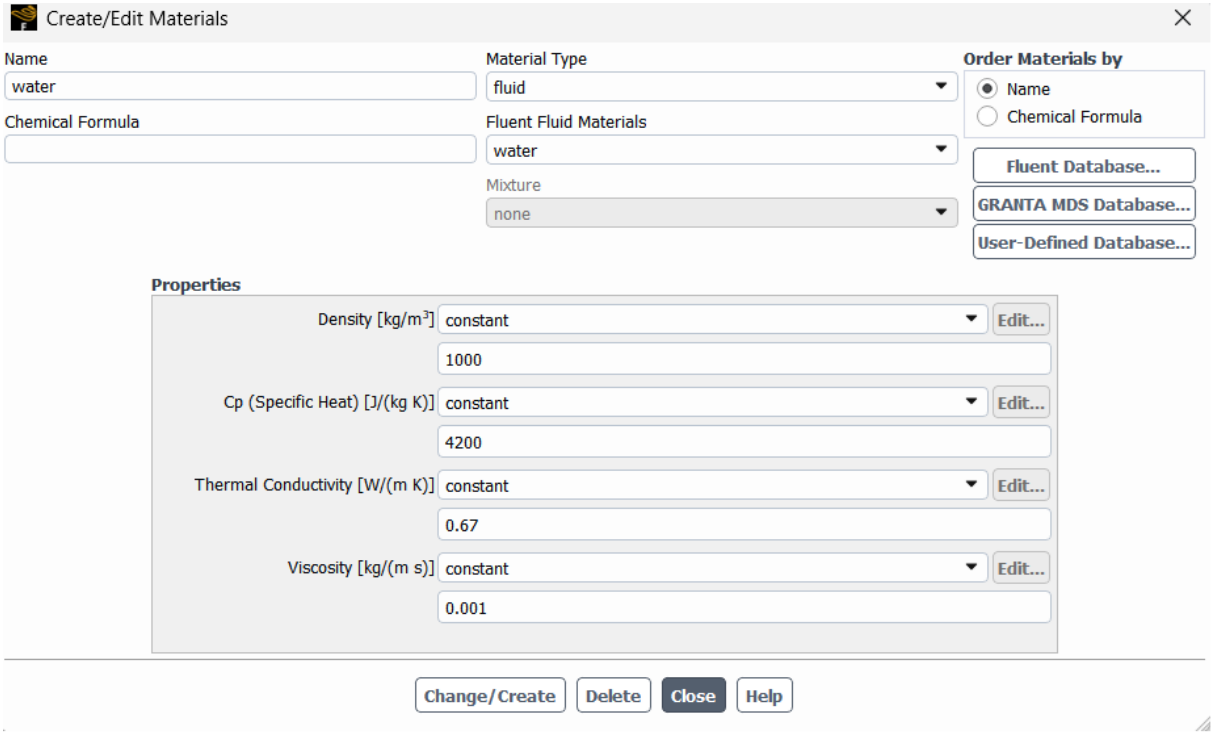


Figure 2. Properties of water used in parts 2 and 3.

The mathematical model relies on the conservation laws of mass, momentum (Navier-Stokes equations), and energy, which are solved numerically using the ANSYS Fluent software. For the turbulent cases, the Reynolds-Averaged Navier-Stokes equations are closed using both the Standard k-epsilon(k-e) and SST k-omega(k-w) turbulence models. The boundary conditions applied include uniform velocity profiles at the inlets, where the large inlet introduces fluid at 23 degrees Celsius (296.15 K) with a velocity of 0.4 m/s (or 1 m/s for specific laminar cases), and the small inlet introduces fluid at 52 degrees Celsius (325.15 K) with velocities varying between 0.8 m/s and 2.4 m/s (Figure 3) to analyze mixing efficiency. A zero-gauge pressure condition is applied at the outlet, and all wall boundaries are treated as stationary with a no-slip condition and zero-gradient heat flux. The flow is assumed to be incompressible, and buoyancy effects are considered negligible throughout the analysis.

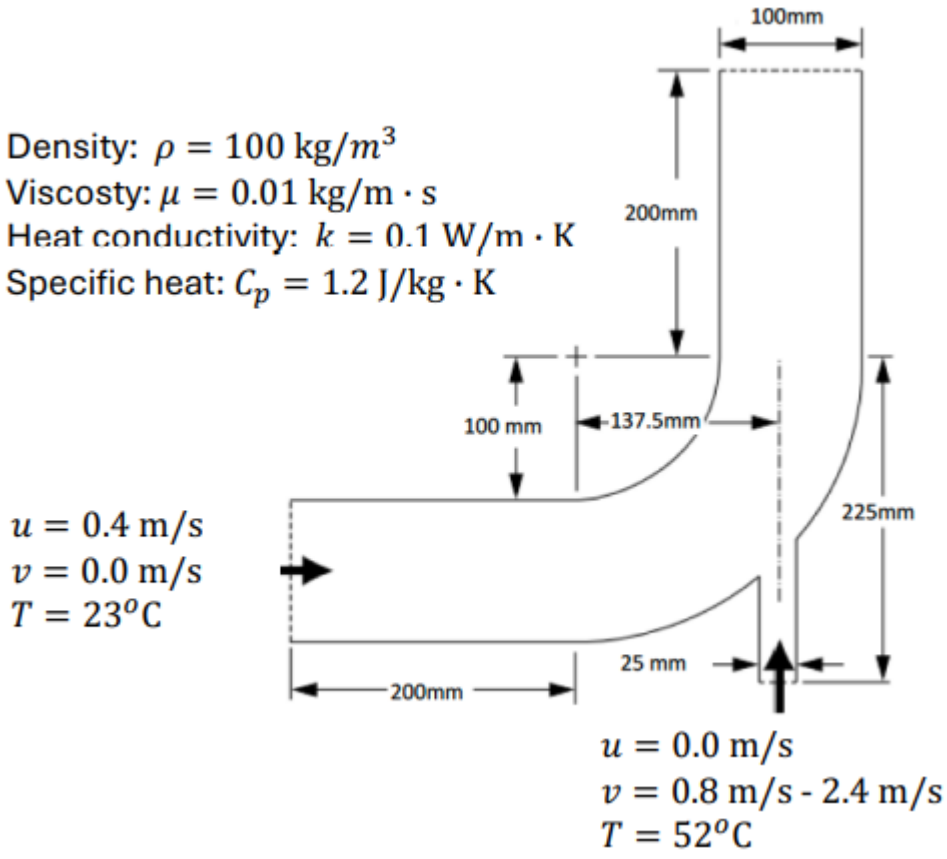


Figure 3. Flow geometry and boundary conditions for Task 1 [1].

Figure 4 and 5 is created using the guidelines and youtube tutorial to initialize the project [1][4].

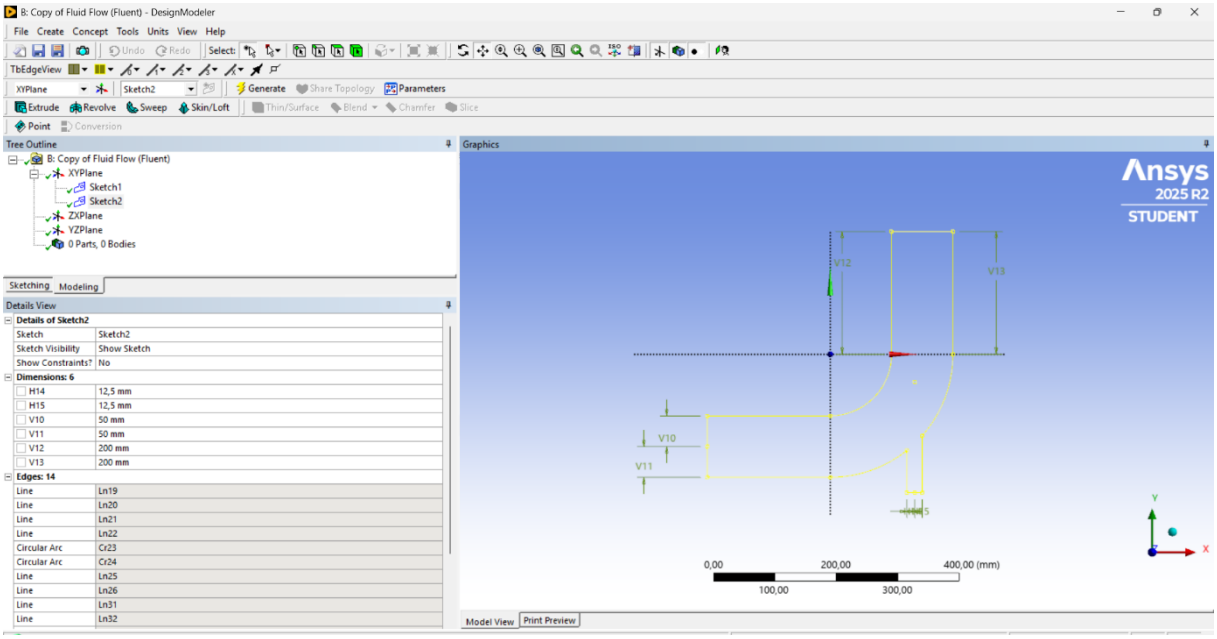


Figure 4, 2D sketching of the model for task 1 and 2. Small inlet and big inlet are perpendicular to each other.

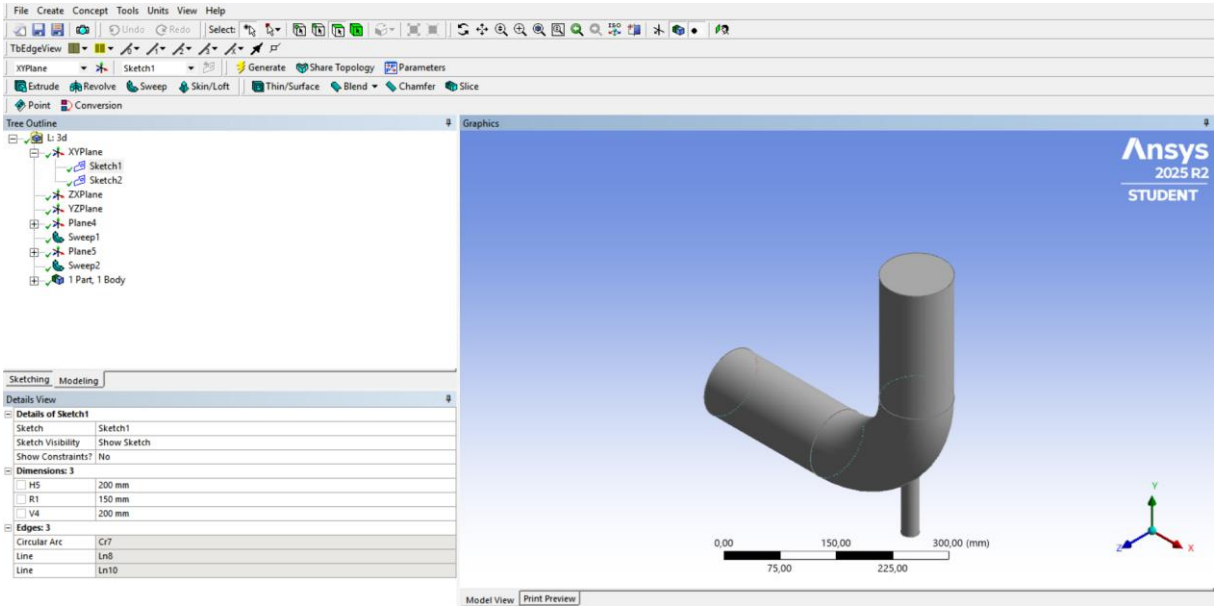


Figure 5. 3D model of part 3. It will be used for understanding how fluid behaves in a 3D environment. This part is drawn with the help of Advanced Engineering Tutorials [4]

2.1. Mathematical Model and Governing Equations

The numerical simulations performed in this study are based on the fundamental conservation laws of fluid mechanics[2]. The flow is assumed to be steady, incompressible, and Newtonian, with buoyancy effects considered negligible [1]. To determine the flow and temperature fields, ANSYS Fluent solves the coupled system of differential equations representing the conservation of mass, momentum, and energy[3].

2.1.1. Conservation of Mass (Continuity Equation)

The continuity equation ensures that mass is conserved within the computational domain. For an incompressible fluid, the divergence of the velocity vector is zero, which implies that the net rate of flow into any control volume is equal to the net rate of flow out.

$$\nabla \cdot \vec{V} = 0$$

Where $\vec{V}$ represents the velocity vector.

2.1.2. Conservation of Momentum

The balance of forces acting on the fluid is described by the Navier-Stokes equations. These equations represent the conservation of momentum, balancing inertial forces with pressure gradients and viscous forces.

$$\rho(\vec{V} \cdot \nabla)\vec{V} = -\nabla p + \mu \nabla^2 \vec{V}$$

Where ρ is the fluid density, p is the static pressure, and μ is the dynamic viscosity.

2.1.3. Conservation of Energy

Since the mixing process involves fluids at different temperatures, the energy equation is solved to predict the heat transfer. This equation balances the transport of energy due to fluid motion (convection) with the energy transfer due to thermal conduction.

$$\rho C_p\left(\vec{V} \cdot \nabla T\right) = k\nabla^2 T$$

Where C_p is the specific heat capacity at constant pressure, T is the temperature, and k is the thermal conductivity.

3. Results and Discussions

3.1. Task 1: Laminar Flow Analysis

In the first phase of the project, the mixing elbow was analyzed under laminar flow conditions ($Re = 400$) using an unstructured triangular mesh (Figure 6). To investigate the effect of momentum ratios on flow dynamics, simulations were performed for three different small inlet velocities: $v = 2\ \mathrm{m/s}, v = 4\ \mathrm{m/s}, and v = 6\ \mathrm{m/s}$.

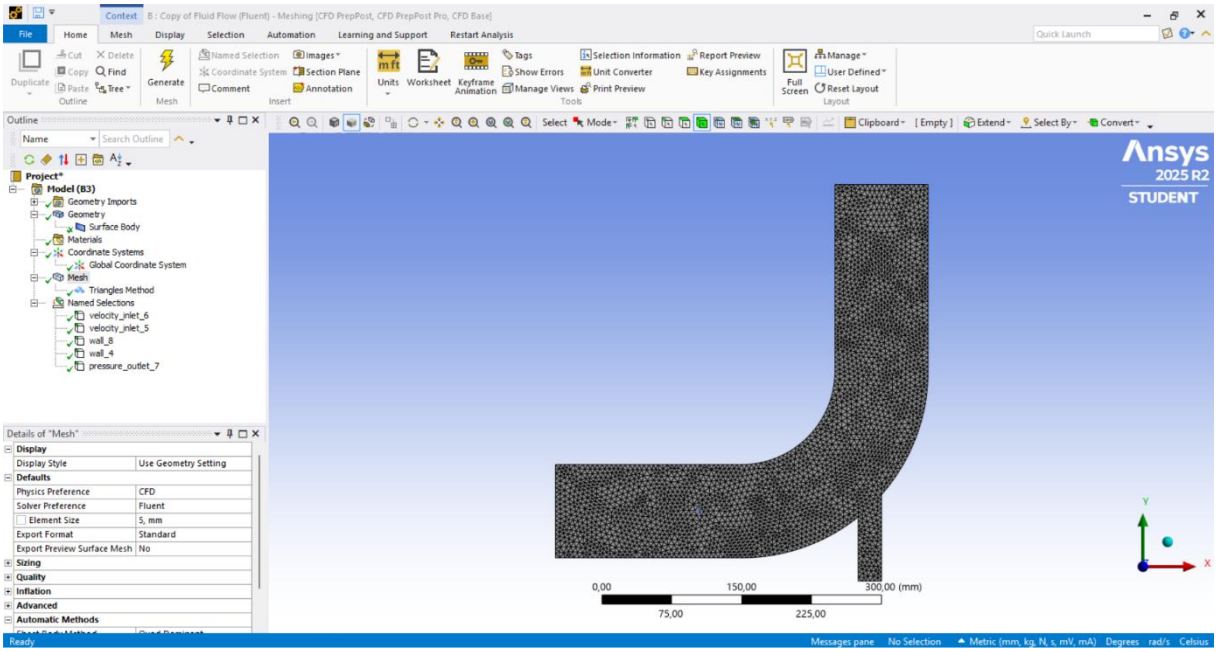


Figure 6. The geometry and unstructured computational grid for task 1.

Figure 7 displays the static pressure distribution at the junction. A high-pressure region is visible on the outer wall of the main pipe, directly opposite the small inlet. This corresponds to the stagnation point where the vertical jet impinges on the wall. The intensity of this high-pressure zone increases with the jet velocity, consistent with the conversion of dynamic pressure to static pressure. Conversely, a low-pressure zone is observed at the sharp corner of the elbow, which drives the flow separation mentioned above.

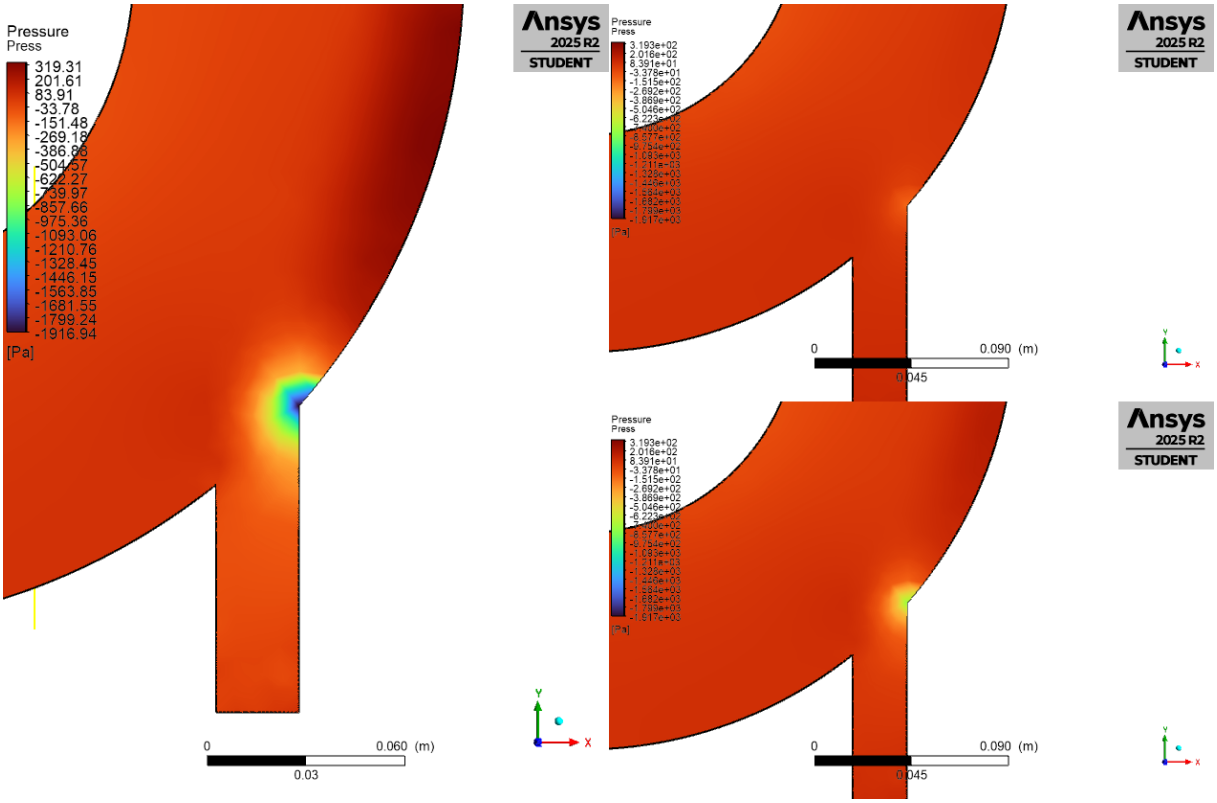


Figure 7. Static pressure contours at the elbow junction. The regions of minimum pressure correspond to the stagnation points where the vertical jet impinges on the channel walls.

The temperature contours in Figure 8 provide insight into the mixing mechanism. Since the flow is laminar, turbulent mixing is absent, and heat transfer is dominated by molecular diffusion and convection.

- Stratification: The contours show distinct layers of hot (red) and cold (blue) fluid flowing side-by-side without significant blending. This phenomenon, known as thermal stratification, indicates poor mixing efficiency in the laminar regime.

- Effect of Velocity: Although true mixing is limited, increasing the jet velocity to $6\ \mathrm{m/s}$ physically displaces the cold fluid and forces the hot stream deeper into the pipe cross-section. This increases the contact interface between the two fluids compared to the $2\ \mathrm{m/s}$ case, where the hot fluid is confined to the wall.

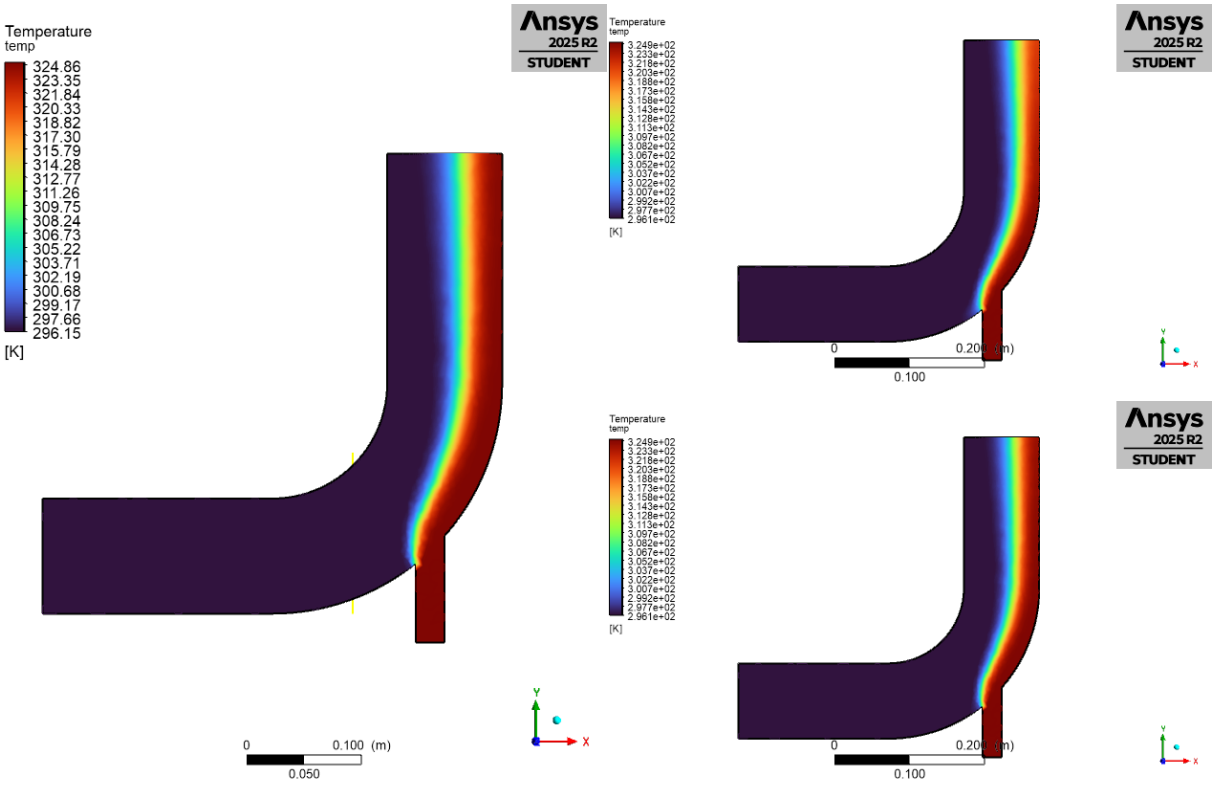


Figure 8. Temperature distribution contours for Task 1 ($v = 2{,}4{,}6\ \mathrm{m/s}$), demonstrating the thermal stratification and the interface between the cold main stream and the hot jet stream.

The velocity magnitude contours presented in Figure 9 illustrate the interaction between the horizontal main flow ($0.4\ \mathrm{m/s}$) and the vertical jet.

- At the lowest velocity ($v = 2\ \mathrm{m/s}$), the vertical jet has insufficient momentum to penetrate deep into the main stream and is quickly deflected downstream, flowing close to the bottom wall. However, as the velocity increases to $6\ \mathrm{m/s}$ (as shown in the leftmost contour of Figure 9), the jet momentum allows it to push further towards the center of the pipe.
- A flow feature observed in all three cases is the low-velocity recirculation zone located at the inner corner of the elbow and immediately downstream of the small inlet pipe. This indicates a flow separation caused by the adverse pressure gradient and the sharp change in geometry.

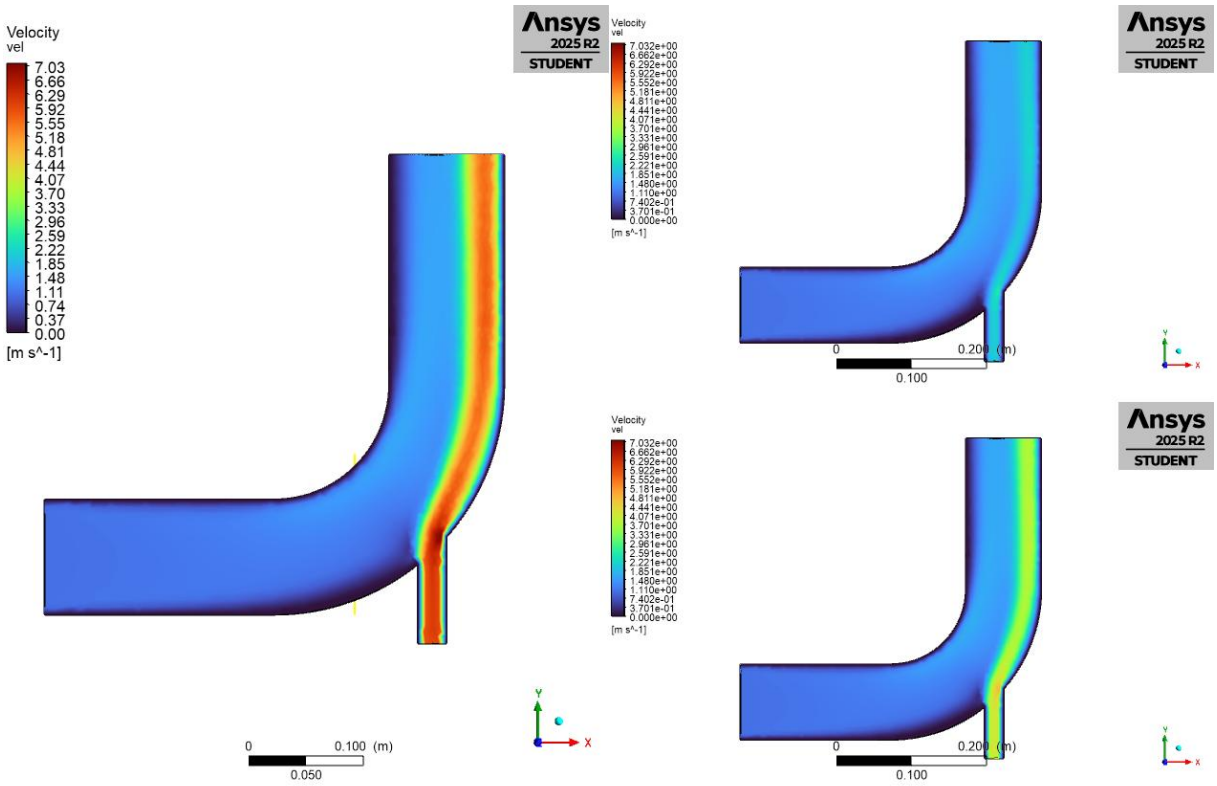


Figure 9. Quantitative profiles taken at cross-sections downstream of the mixing elbow: Velocity profiles showing the variation in flow development for different inlet velocities ($v = 2{,}4{,}6\ \mathrm{m/s}$).

The flow physics observed qualitatively in the contour plots are rigorously substantiated by the quantitative profiles extracted at the pipe outlet, as presented in Figure 7,8 and 9. The velocity magnitude profiles reveal a highly asymmetric distribution, confirming that the flow has not fully developed into a standard parabolic profile due to the strong influence of the secondary jet. Specifically, the case with the highest inlet velocity ($v = 6\ \mathrm{m/s}$, red line) exhibits a pronounced high-momentum peak near the outer wall, indicating that the jet core preserves its kinetic energy and resists deflection by the main stream. In contrast, the lowest velocity case ($v = 2\ \mathrm{m/s}$, blue line) shows a significantly suppressed peak, suggesting rapid momentum dissipation. More critically, the temperature profiles in Figure 10 is indicator of mixing efficiency. Rather than converging to a uniform, average temperature (which would appear as a horizontal line in a well-mixed scenario), the profiles display a steep gradient where the fluid transitions abruptly from the cold mainstream temperature ($296.15\ \mathrm{K}$) to the hot jet temperature ($325.15\ \mathrm{K}$). While increasing the jet velocity to $6\ \mathrm{m/s}$ shifts the thermal interface towards the center of the pipe effectively displacing the cold fluid volume, the preservation of this sharp temperature gradient proves that the two streams remain

thermally stratified. This confirms that in the laminar regime, heat transfer is governed almost exclusively by convection along streamlines with negligible diffusive mixing across the interface.

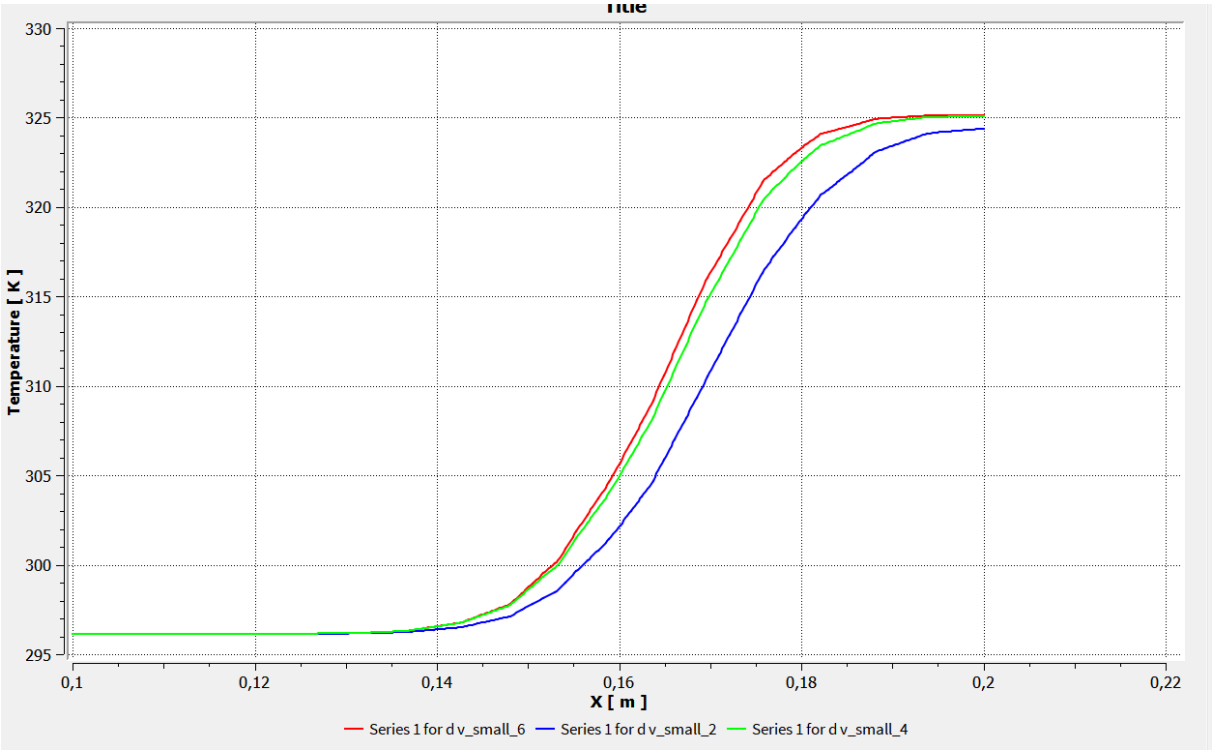


Figure 10. Temprature profiles at the outlet section for three different small inlet velocities $v = 2{,}4{,}6\ \mathrm{m/s}$ after 2 inlets mixes.

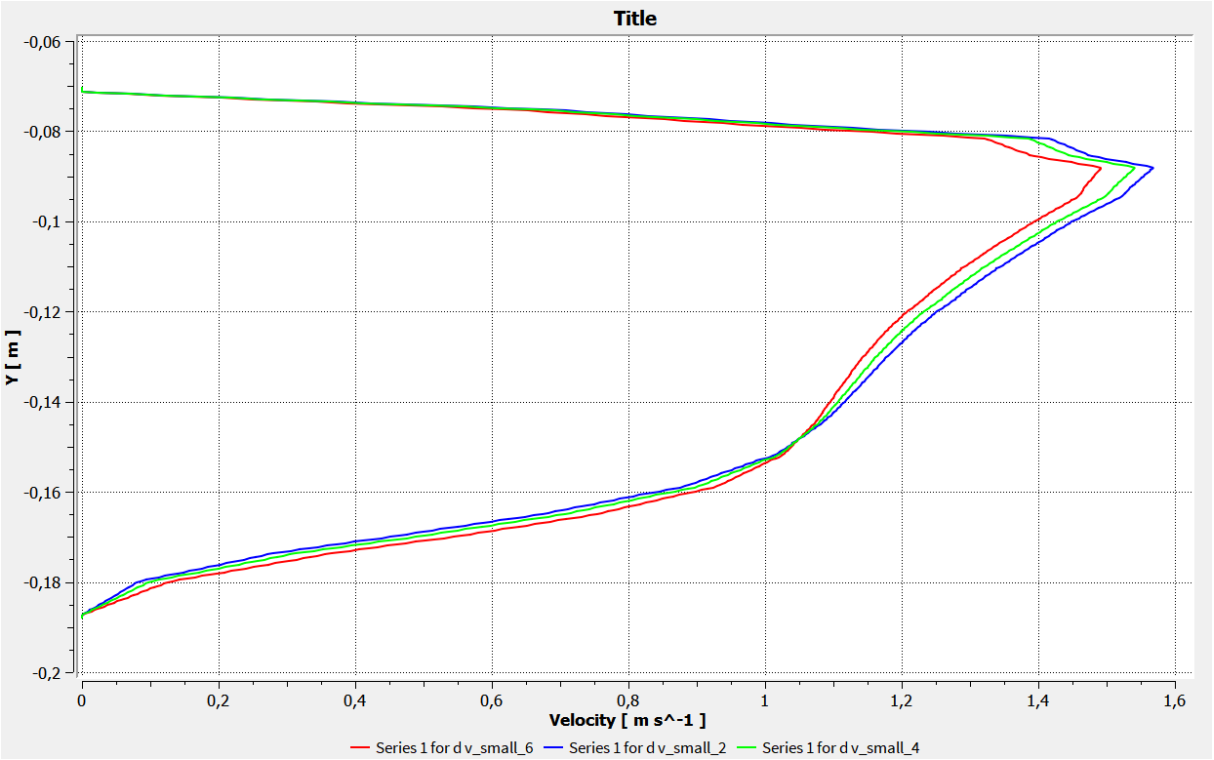


Figure 11. Velocity profiles at the inlet section for three different small inlet velocities $v = 2{,}4{,}6$ m/s. Before 2 inlets mixes.

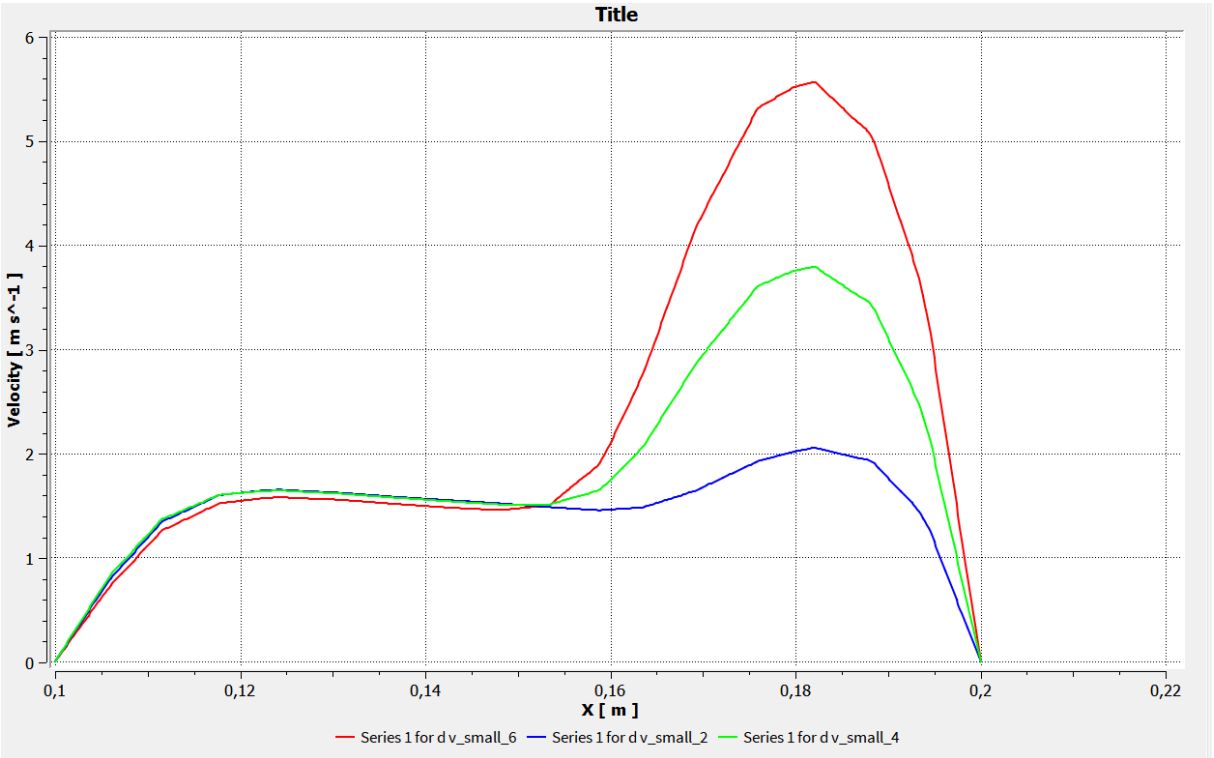


Figure 12. Velocity profiles at the outlet section for three different small inlet velocities $v = 2{,}4{,}6$ m/s. after 2 inlets mixes.

3.2. Task 2: Turbulent Flow Analysis

Following the fundamental investigation of laminar flow in Task 1, the second phase of this project focuses on the more complex and industrially relevant regime of turbulent flow. In this section, the working fluid is changed to water, and the Reynolds number is increased to simulate high momentum mixing processes commonly encountered in engineering applications.

The analysis is structured into three consecutive stages to ensure numerical accuracy and physical insight:

1. Grid Independence Study: First, the reliability of the numerical solution is verified by performing simulations on three distinct structured grids (Coarse, Medium, and Fine) to prove that the results are independent of mesh density.
2. Turbulence Model Comparison: Second, a sensitivity analysis is conducted by comparing the results obtained from the Standard $k-\epsilon$ and SST $k-\omega$ models to demonstrate that the chosen turbulence approximation does not introduce significant deviations in the flow field prediction.
3. Parametric Study: Finally, using the verified grid and the appropriate turbulence model, the effect of jet momentum on mixing efficiency is analyzed by varying the small inlet velocity ($v = 1,2, \text{and } 3$ m/s).

3.2.1. Mesh Convergance

In order to verify the reliability of the numerical results and ensure that the solution is independent of the spatial discretization, a systematic grid convergence study was conducted. For this purpose, three distinct structured computational grids were generated using the Face Meshing method with quadrilateral elements to better resolve the boundary layer flows in the turbulent regime. The grids were created by varying the element size to 8 mm$(Coarse), 6$ mm$(Medium), and 4$ mm (Fine). As illustrated in Figure 13, this refinement process resulted in total node counts of 1148, 1954, and 4323, respectively.

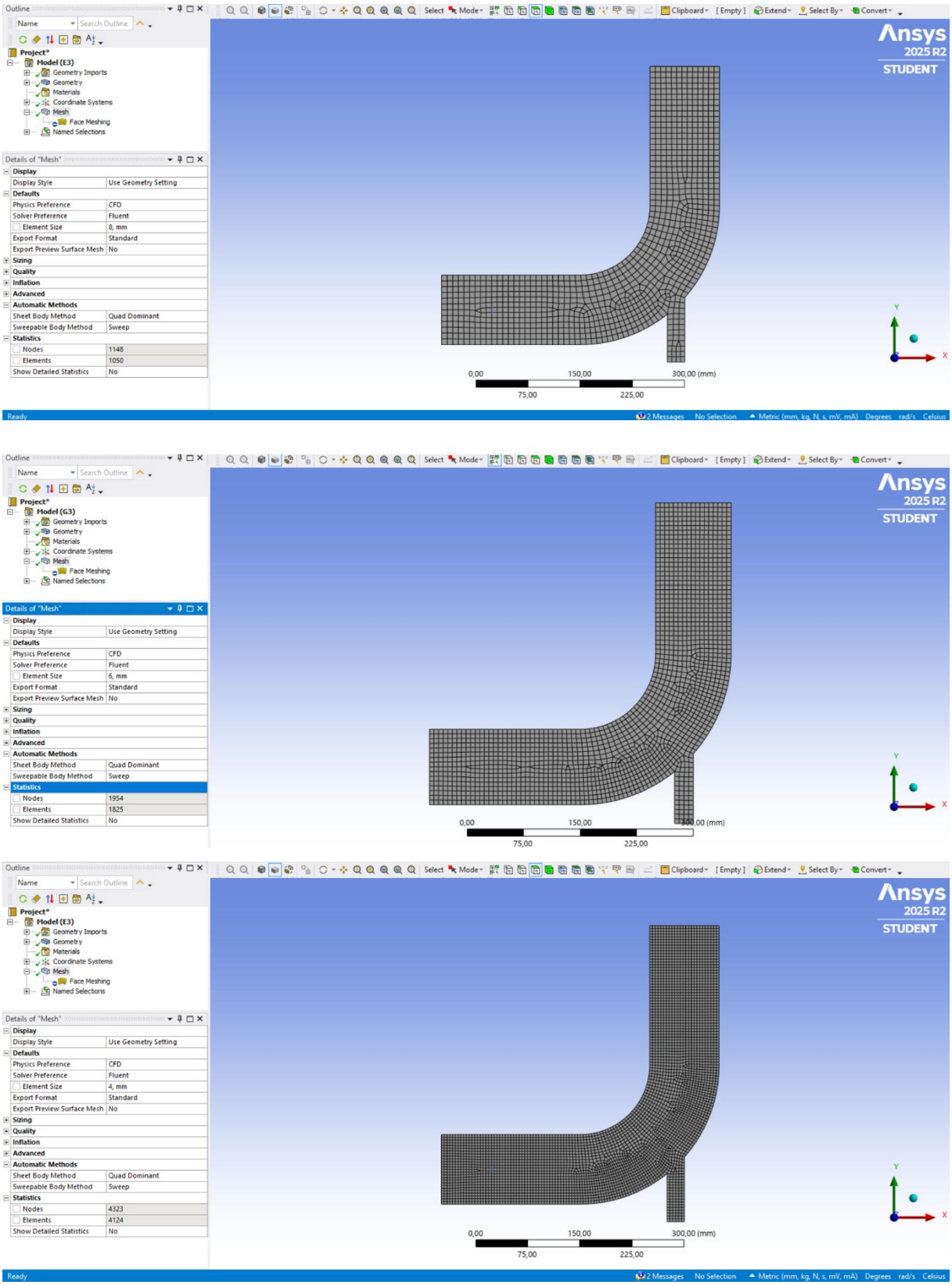


Figure 13. Structured computational grids used for the grid independence study: (a) Coarse mesh with $8\ \mathrm{mm}$ element size (1148 nodes), (b) Medium mesh with $6\ \mathrm{mm}$ element size (1954 nodes), and (c) Fine mesh with $4\ \mathrm{mm}$ element size (4323 nodes)

To quantitatively evaluate the convergence, the velocity magnitude profiles were extracted at a specific cross-section downstream of the mixing elbow for all three mesh cases. As shown in Figure 14, the results demonstrate a clear trend towards convergence.

The profile obtained from the Coarse Mesh (Blue line, $8\ \mathrm{mm}$) shows a noticeable deviation in peak velocity and profile shape compared to the finer meshes.

As the mesh is refined to the Medium Mesh (Red line, $6\ \mathrm{mm}$) and subsequently to the Fine Mesh ($Green\ line, 4\ mm$), the variation between the profiles diminishes.

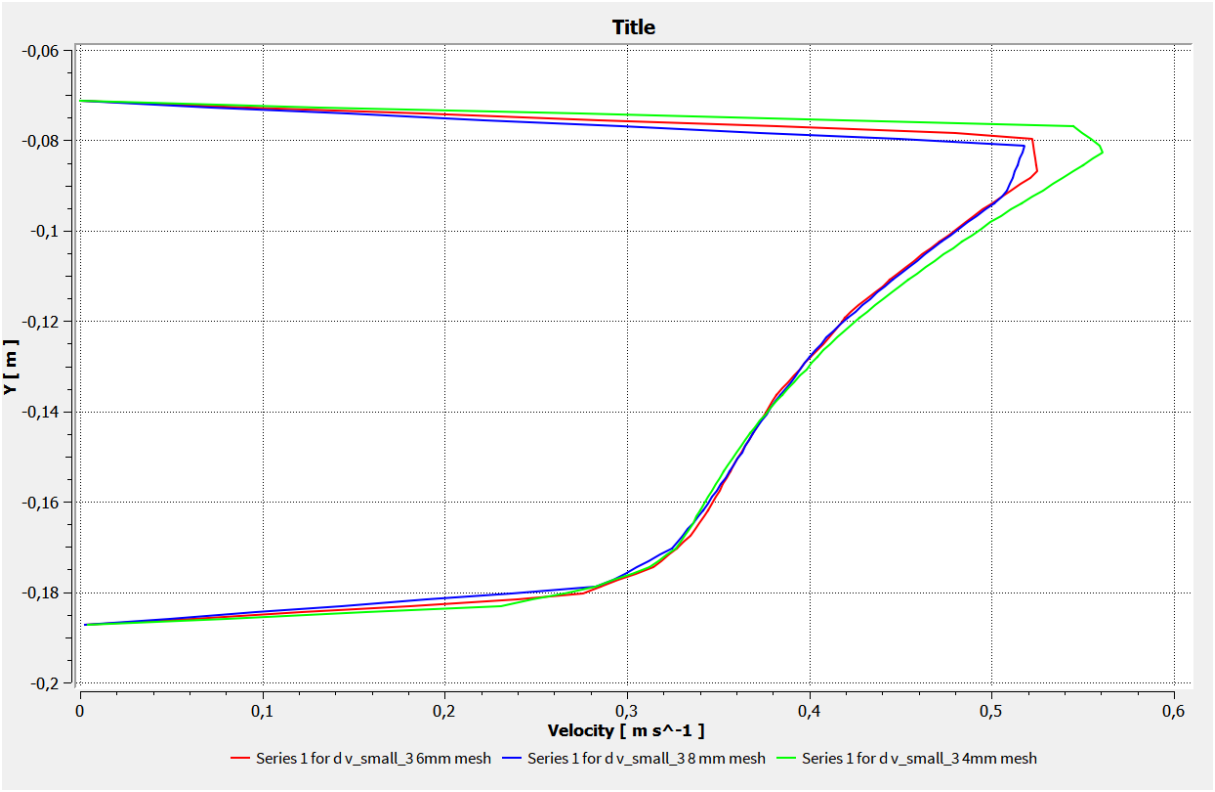


Figure 14. Grid convergence analysis comparing the velocity magnitude profiles at the outlet section for three different mesh sizes:$Coarse(8\ \mathrm{mm}, Blue), Medium(6\ \mathrm{mm}, Red), andFine(4\ \mathrm{mm}, Green)$. The reducing gap between the curves indicates the approach towards a grid-independent solution.

In addition to the line plots, the temperature field was visually compared across the three mesh densities as presented in Figure 15. The thermal mixing patterns, specifically the penetration of the hot jet into the cold stream and the development of the thermal boundary layer along the outer

wall, are qualitatively identical between the Medium and Fine meshes.

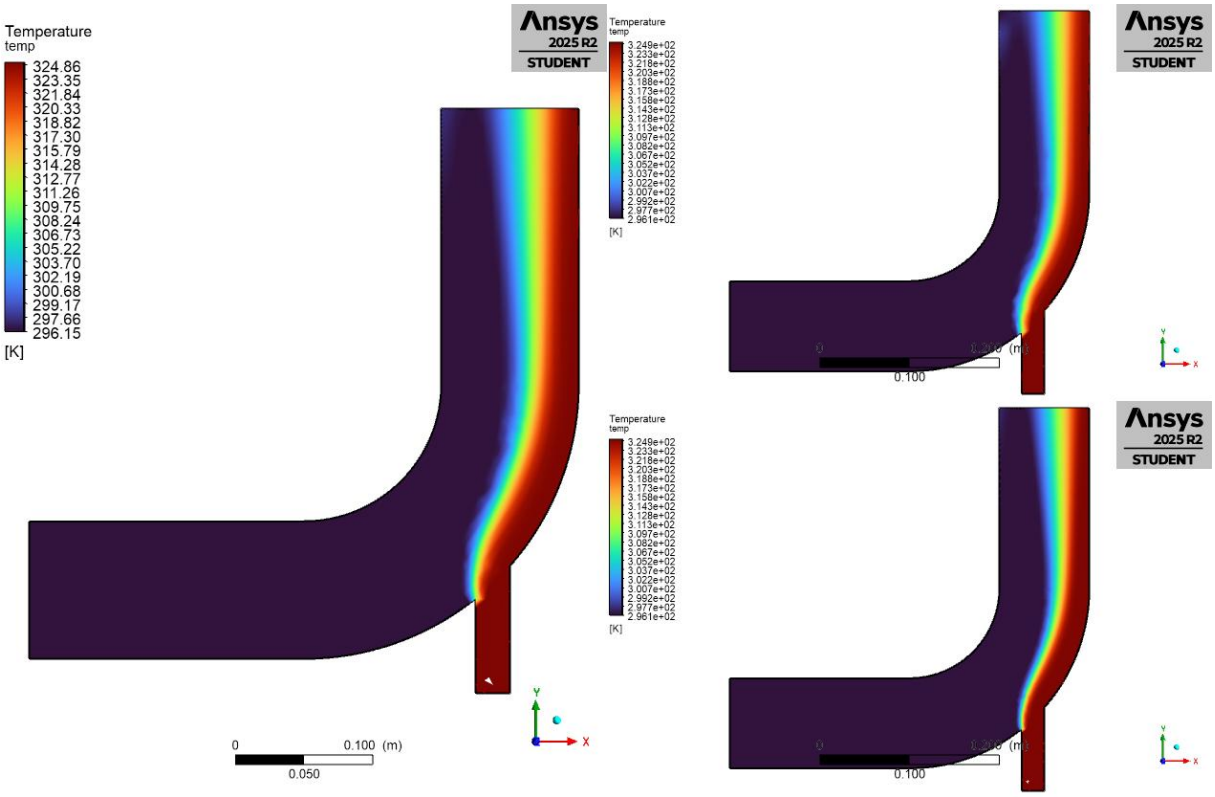


Figure 15. Qualitative comparison of temperature contours for the Coarse, Medium, and Fine meshes. The visual similarity between the medium and fine mesh results supports the quantitative evidence of grid convergence.

3.2.2. Comparison of Turbulence Models

To evaluate the sensitivity of the numerical results to the turbulence modeling approximation, the simulation was performed using two distinct Reynolds-Averaged Navier-Stokes (RANS) closure models: the Standard $k-\epsilon$ model and the SST $k-\omega$ model. The Standard $k-\epsilon$ model is widely used for industrial flows due to its robustness, while the SST (Shear Stress Transport) $k-\omega$ model is known for its superior performance in resolving flow separation and wall physics.

The velocity magnitude profiles extracted from the outlet section for both models are shown in Figure 16. As clearly illustrated in the graph, the results obtained from both models are nearly identical. The velocity profiles overlap significantly throughout the pipe cross-section, capturing the same peak velocity location and magnitude. Only negligible discrepancies are observed near the wall regions and the peak inflection point. This similarity suggests that for the current Reynolds number and geometry, the bulk flow features are not heavily dependent on the specific turbulence formulation.

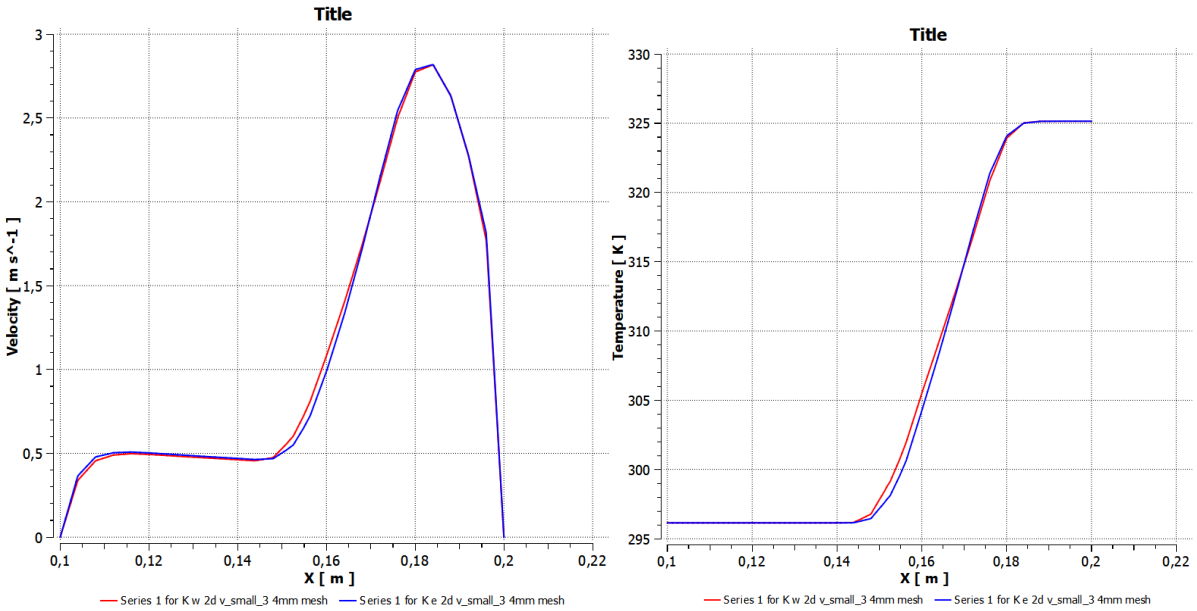


Figure 16. Comparison of velocity magnitude profiles and temprature values at the outlet section obtained using two different turbulence models: Standard $k-\epsilon$ (Blue line) and SST $k-\omega$ (Red line). The close agreement between the two profiles indicates that the solution is insensitive to the choice of turbulence model for this specific flow regime.

3.2.3. Turbulent Flow Analysis

The turbulent flow field within the mixing elbow was analyzed using the Standard $k-\epsilon$ turbulence model on a structured grid, considering water as the working fluid. The analysis is divided into qualitative assessments of the flow field and quantitative parametric studies to evaluate mixing efficiency.

The qualitative flow patterns, visualized through streamlines in Figure 17, reveal hydrodynamic features characteristic of high Reynolds number flows. A flow separation is observed on the inner wall immediately downstream of the elbow. This separation is induced by the adverse pressure gradient created as the flow attempts to follow the sharp curvature.

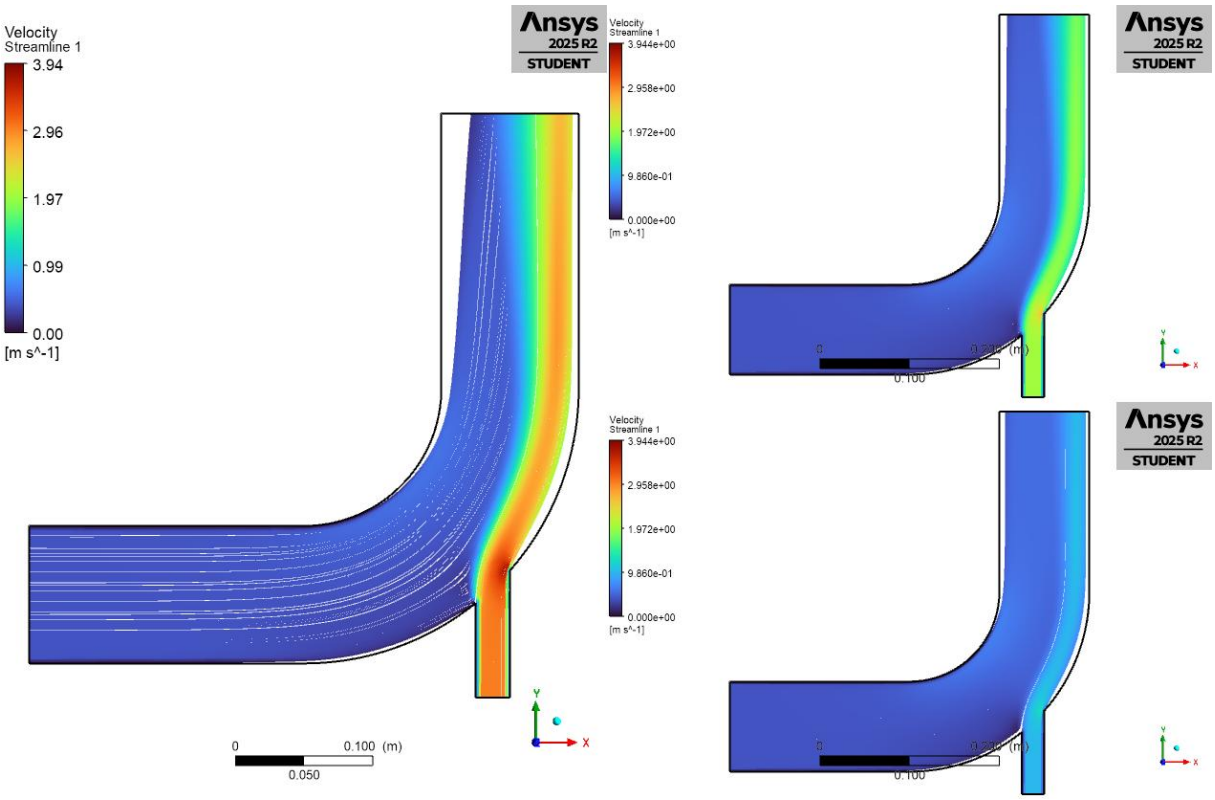


Figure 17. Velocity streamlines colored by velocity magnitude. The visualization clearly highlights the turbulent mixing region and the formation of a recirculation zone (flow separation) immediately downstream of the junction on the inner wall. ($v = 1,2, \text{and } 3\ \text{m/s}$)

The static pressure distribution presented in Figure 18 shows the forces acting on the walls. A distinct high pressure stagnation region is observed on the outer wall where the vertical jet impinges, converting dynamic pressure into static pressure. Conversely, a low-pressure zone forms at the sharp inner corner, driving the aforementioned separation. It is important to note that the pressure contours are displayed using a logarithmic scale to clearly visualize the wide range of pressure gradients occurring within the mixing zone.

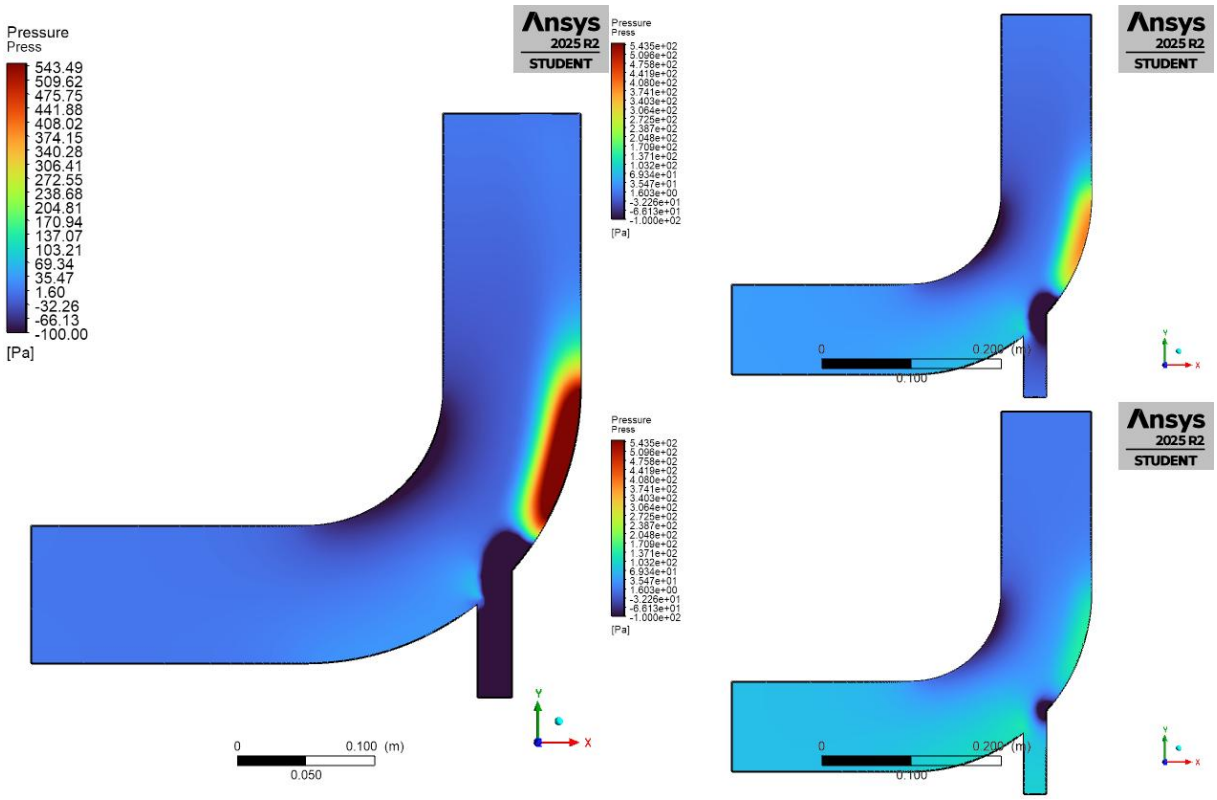


Figure 18. Static pressure distribution contour for the turbulent flow case. A distinct high-pressure stagnation region is observed on the outer wall where the vertical jet impinges, while a low-pressure zone forms at the sharp corner of the elbow. Also note that the scale is logoritmic. ($v = 1,2, \text{and } 3\ \text{m/s}$)

The temperature distribution contours in Figure 19 present a fundamental departure from the laminar case. Unlike the distinct stratification seen previously, the thermal interface between the cold mainstream and the hot vertical jet appears diffuse and chaotic. This indicates that turbulent eddies significantly enhance the effective thermal diffusivity, allowing for rapid energy exchange between the fluid layers before they reach the outlet.

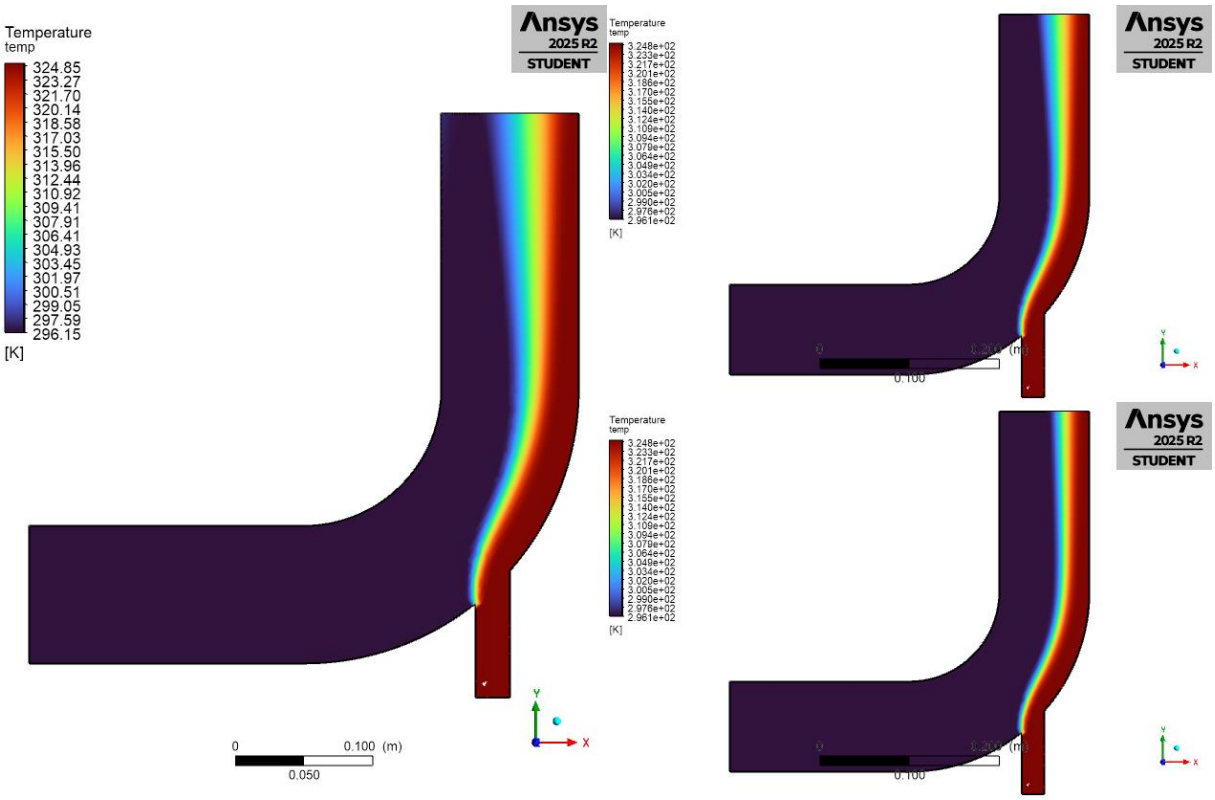


Figure 19. Temperature distribution contour showing the interaction between the cold main stream and the hot vertical jet. Compared to the laminar case, the thermal interface appears more diffuse, indicating enhanced mixing due to turbulent eddies. ($v = 1$,2, and 3 m/s)

To quantify the mixing performance, a parametric study was conducted by varying the small inlet velocity ($v = 1$,2, and 3 m/s).

- Outlet Velocity Profiles: The impact of the jet momentum is clearly visible in the outlet velocity profiles presented in Figure 20. As the inlet velocity increases to 3 m/s, the peak velocity magnitude shifts significantly towards the outer wall. This confirms that the high-momentum jet penetrates the cross-flow effectively, altering the fully developed profile expected in a straight pipe.
- Mixing Efficiency: Most critically, the temperature profiles extracted at the outlet (Figure 21) demonstrate a positive correlation between inlet velocity and mixing quality. At the lowest velocity ($v = 1$ m/s), the hot fluid remains confined near the wall. However, as the velocity increases to 3 m/s, the generated turbulent fluctuations vigorously mix the streams. Consequently, the temperature profile for the 3 m/s case is significantly flatter and smoother compared to the steep gradients of the lower velocity cases, indicating a superior and more homogeneous thermal field.

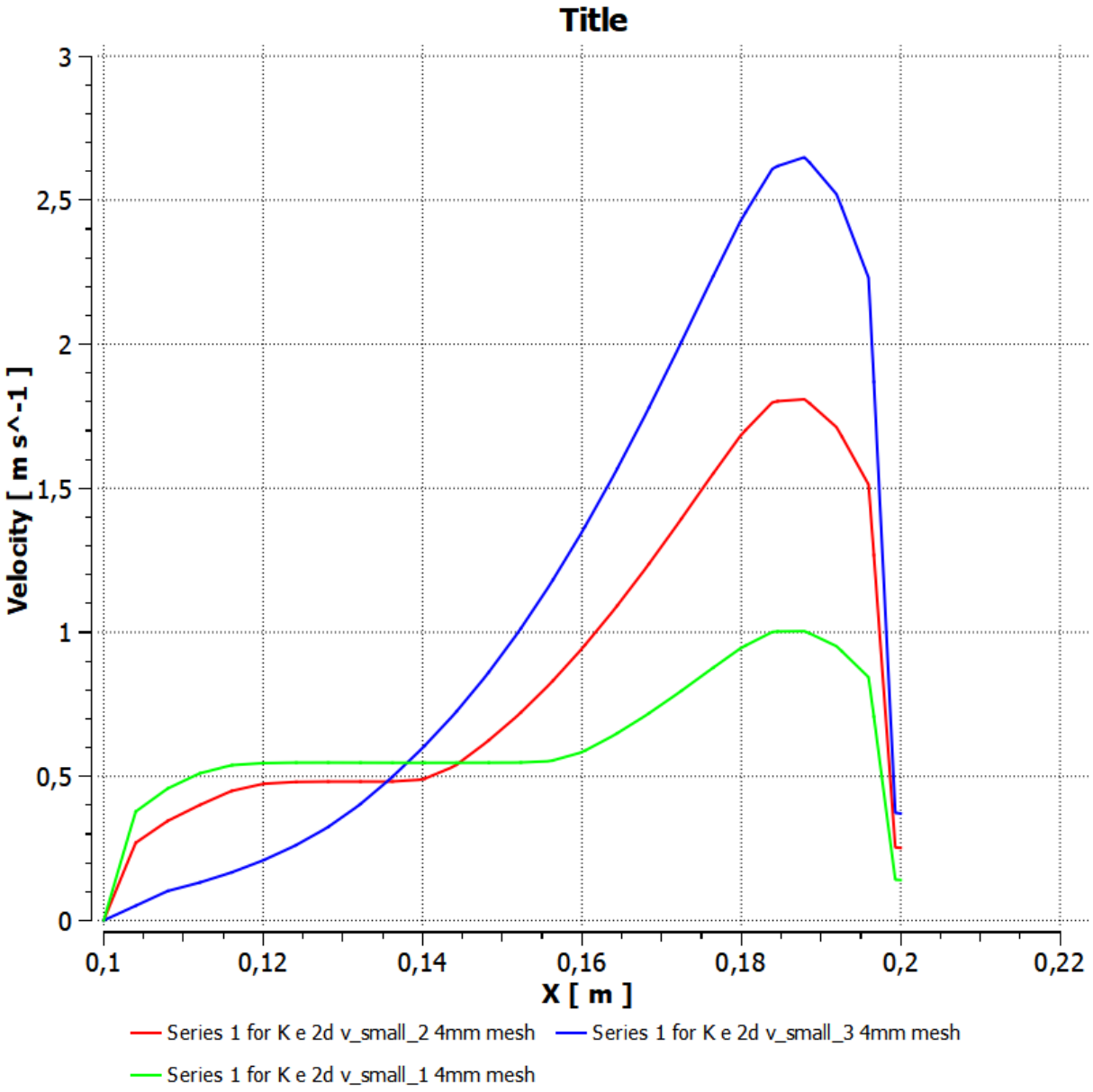


Figure 20. Comparison of velocity magnitude profiles at the outlet section for three different small inlet velocities ($v = 1,2, \text{and } 3\ \text{m/s}$). The peak velocity magnitude shifts towards the outer wall as the jet momentum increases, significantly altering the flow structure.

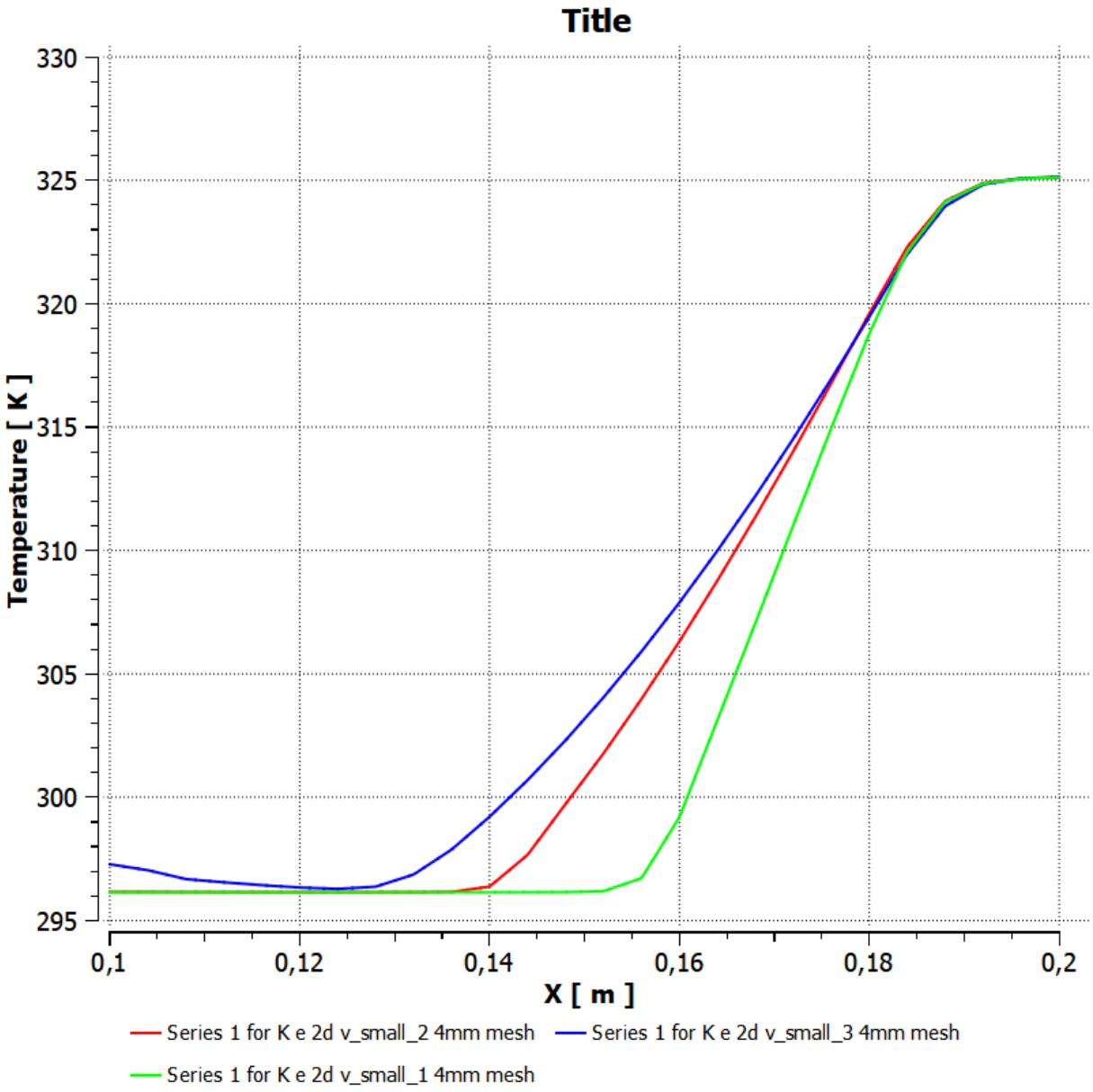


Figure 21. Temperature profiles extracted at the outlet section for varying inlet velocities ($v = 1,2, \text{and } 3\ \text{m/s}$). Unlike the steep gradients observed in the laminar case, these profiles exhibit a smoother transition between the temperature extremes, demonstrating the superior mixing efficiency of the turbulent regime.

3.3. Task 3: Three-Dimensional Flow and Parametric Analysis

In the final phase of the project, the mixing elbow was modeled as a full three-dimensional domain with circular pipe cross-sections. Unlike the 2D planar assumption, the 3D geometry captures critical flow features such as secondary flows (Dean vortices) induced by the centrifugal forces as the fluid navigates the elbow curvature. As visualized in the streamline contours in Figure 22, these vortical

structures cause the fluid to swirl, promoting lateral mixing that cannot be predicted in 2D simulations.

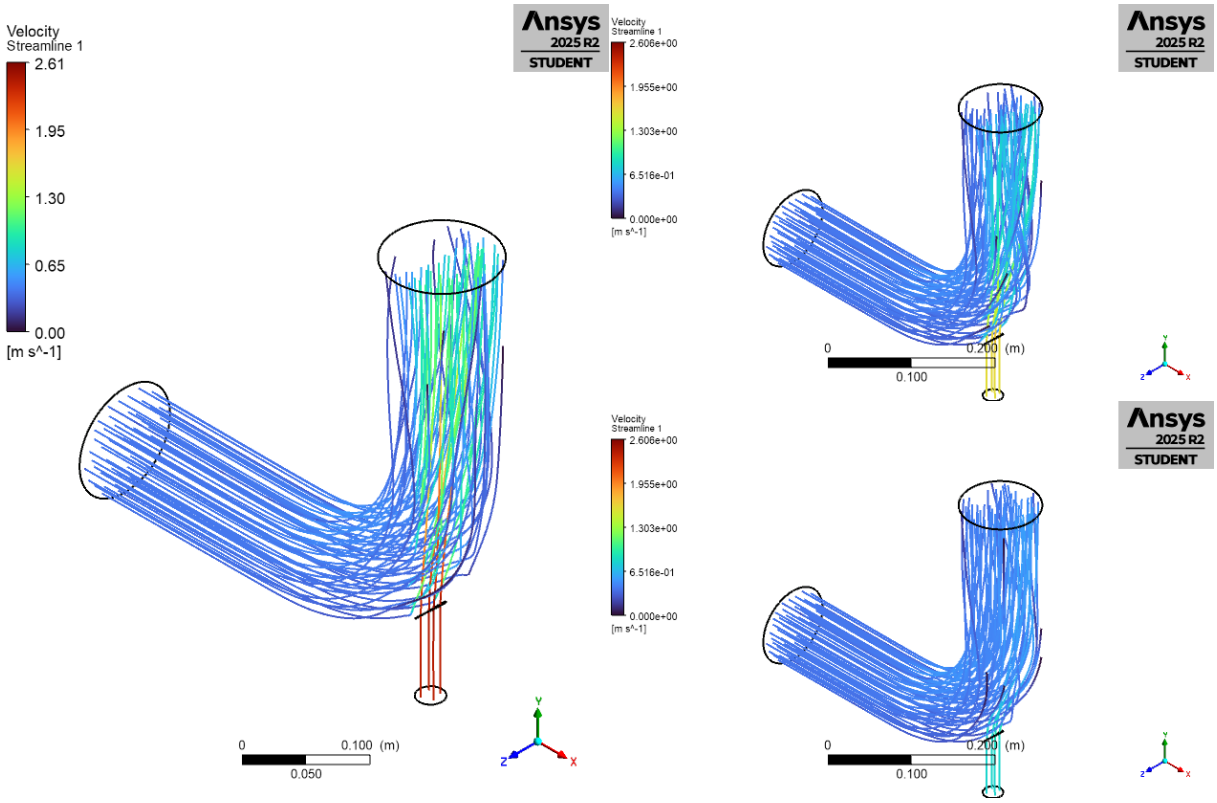


Figure 22. Three-dimensional velocity streamlines originating specifically from the small vertical inlet $v = 0.8, 1.6, \text{and } 2.4\ \text{m/s}$, colored by velocity magnitude. This visualization highlights the trajectory and penetration of the high-momentum jet into the cross-flow, revealing complex three-dimensional flow structures that cannot be captured in 2D planar simulations.

To evaluate the mixing performance in this realistic 3D domain, a parametric study was conducted by varying the small inlet velocity ($v = 0.8, 1.6, \text{and } 2.4\ \text{m/s}$). The temperature profiles extracted at the outlet centerline, presented in Figure 23 quantify the effect of increasing jet momentum.

Low Velocity ($0.8\ \text{m/s}$): At the lowest velocity, the thermal profile indicates that the hot fluid is still relatively confined. The lack of sufficient momentum prevents the jet from penetrating deep into the cross-flow, resulting in a thermally stratified outlet where the hot and cold zones are distinct.

High Velocity ($2.4\ \text{m/s}$): As the velocity increases to $2.4\ \text{m/s}$, the profile shape changes. The higher momentum flux allows the vertical jet to penetrate effectively into the core of the main stream. Consequently, the temperature distribution becomes broader and more widely spread across

the pipe diameter, rather than being confined to a narrow peak near the wall. This indicates that higher velocities intensify the turbulent mixing, leading to a more homogeneous thermal field at the outlet.

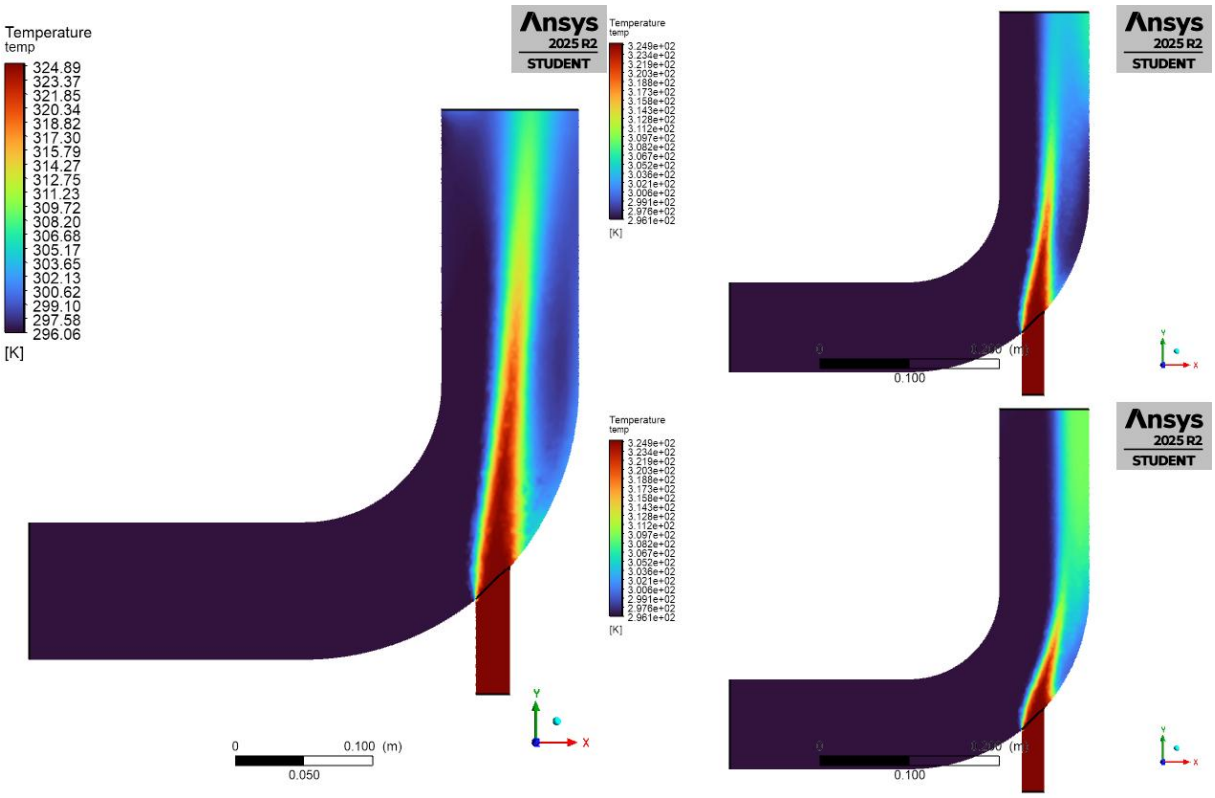


Figure 23. Temperature contours taken at the mid-span symmetry plane ($Z = 0$) of the 3D geometry. The thermal patterns observed in this 2D slice closely match the results obtained from the 2D planar simulations in Task 2.

To characterize the flow state at the exit of the computational domain, the static pressure distribution across the whole body is presented in Figure 24. Although a uniform gauge pressure is applied as a boundary condition, the contour reveals slight non-uniformities and radial gradients. These variations are indicative of the residual swirling motion and the complex secondary flows (Dean vortices) that persist downstream of the elbow.

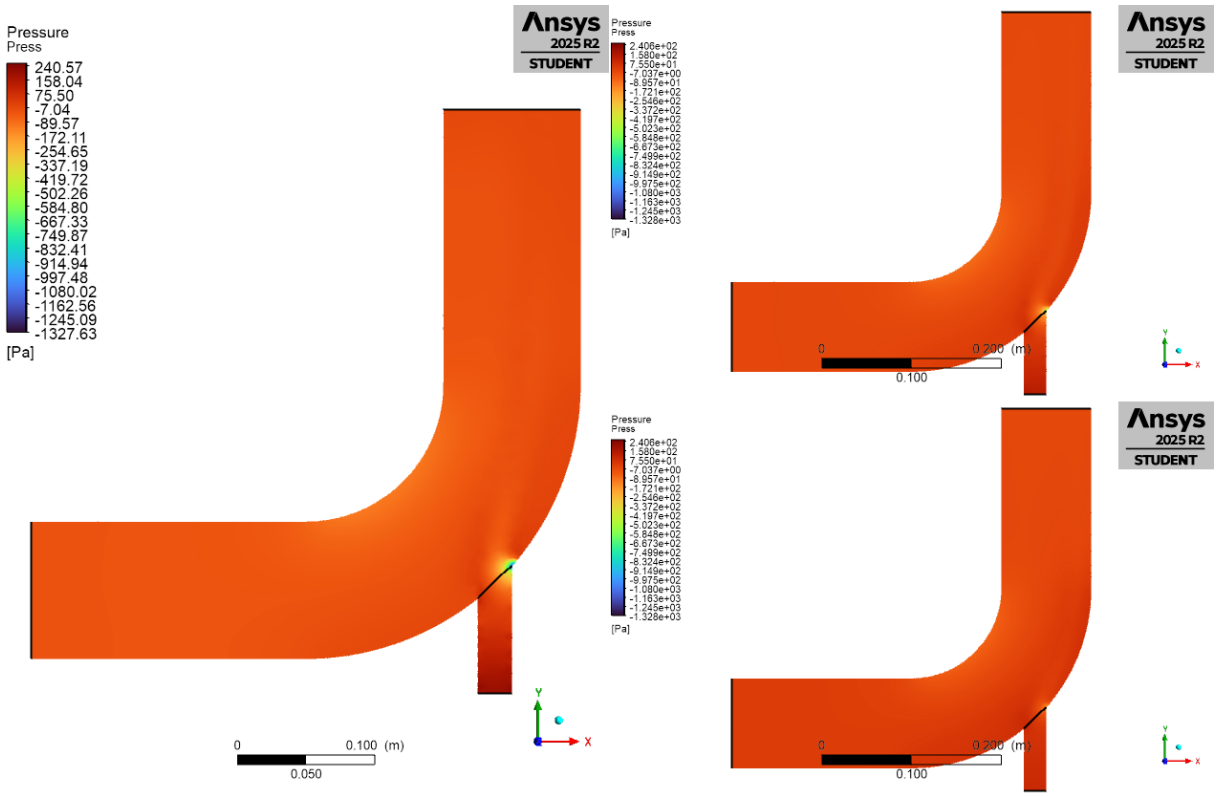


Figure 24. Pressure distribution contour on the circular outlet cross-section of the 3D mixing elbow. $v = 0.8, 1.6,$ and 2.4 m/s

Finally, the quantitative mixing performance is summarized in Figure 25, which plots the temperature profiles along the vertical centerline of the outlet for the three different inlet velocities ($v = 0.8, 1.6,$ and 2.4 m/s). At the lowest velocity (0.8 m/s), the temperature profile is heavily skewed towards the wall, indicating poor penetration and limited mixing. As the velocity increases to 2.4 m/s, the thermal peak shifts towards the center of the pipe, and the distribution becomes broader. Unlike the sharp "step" gradients seen in 2D simulations, these smooth, bell shaped 3D profiles prove that the vertical jet is not just displacing the fluid but is being actively dispersed by the three dimensional vortical structures, leading to a more homogenized mixture.

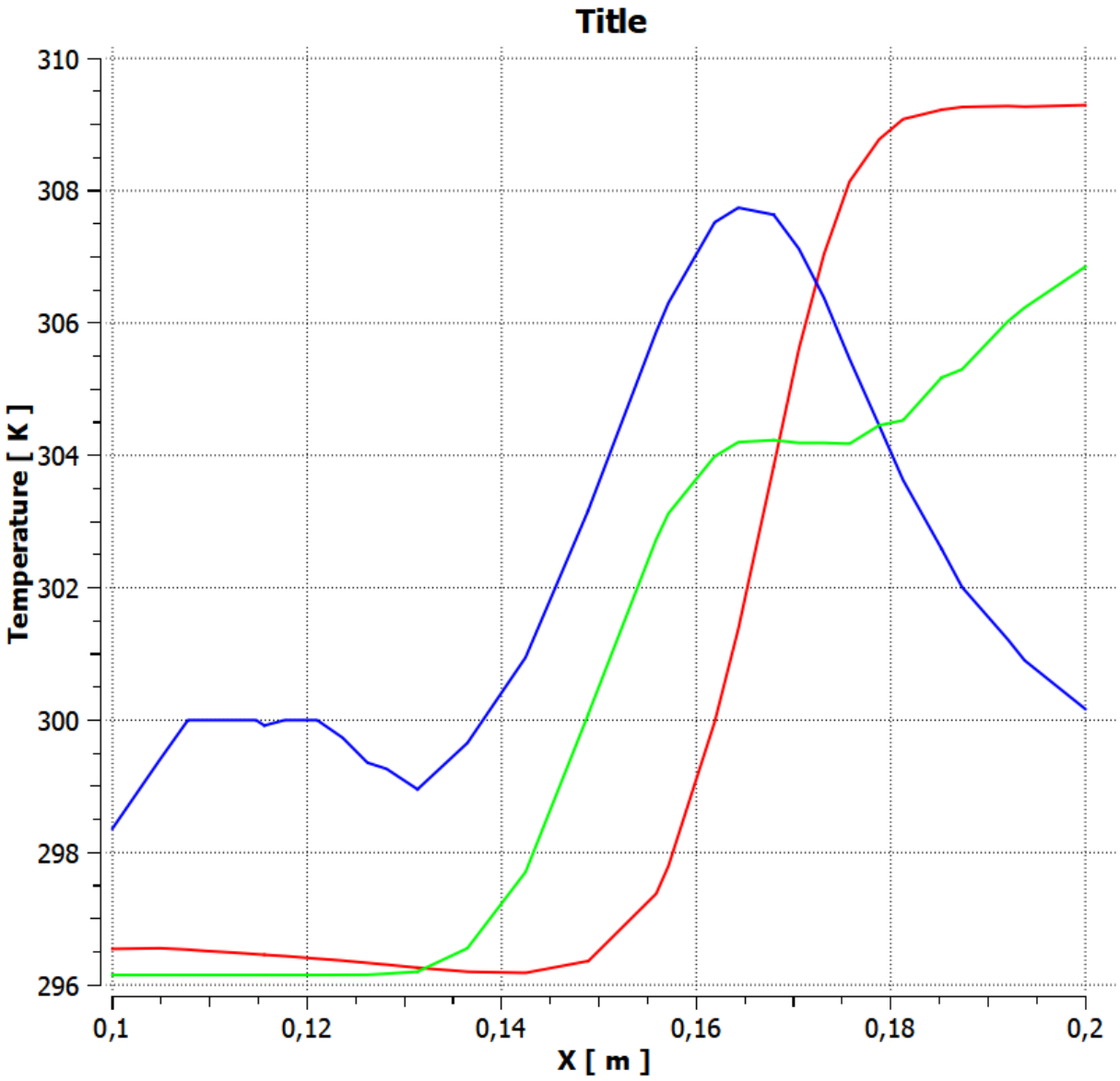


Figure 25. Temperature profiles extracted along the vertical centerline of the outlet section for the 3D circular pipe geometry at three different small inlet velocities ($v = 0.8\ red$,$1.6\ green$, and 2.4 m/s blue). The variations in the profile shape demonstrate the influence of jet momentum on the thermal distribution and mixing quality within the three-dimensional domain.

4. Conclusion

In this comprehensive computational fluid dynamics (CFD) study, the flow and heat transfer characteristics within a mixing elbow were analyzed across three distinct phases, evolving from a basic 2D laminar model to a realistic 3D turbulent simulation. In the laminar flow analysis (Task 1), the results demonstrated that mixing is predominantly governed by convection with negligible diffusion. The flow exhibited significant thermal stratification, where the hot and cold streams flowed side-by-

side with minimal interaction. A recirculation zone was successfully captured downstream of the elbow, highlighting the impact of adverse pressure gradients in curved geometries. The transition to turbulent flow (Task 2) marked a significant improvement in mixing efficiency. The grid independence study confirmed that the fine structured mesh ($4\ \mathrm{mm}$) provided a numerically reliable solution. The comparison between the Standard $k-\epsilon$ and SST $k-\omega$ turbulence models revealed that both models yielded nearly identical velocity profiles, proving the solution's insensitivity to the specific turbulence closure for this bulk flow regime. The parametric study further established a direct positive correlation between the inlet velocity and mixing quality; higher momentum fluxes not only allowed for deeper jet penetration but also enhanced the turbulent diffusivity, resulting in a more homogeneous thermal field at the outlet. Finally, the 3D analysis (Task 3) provided the most physically accurate representation of the problem. Unlike the 2D planar assumption, the 3D simulation captured the curvature-induced secondary flows (Dean vortices). These vortical structures were found to be critical in promoting lateral mixing, demonstrating that 2D simplifications significantly underpredict the mixing performance in such geometries.

5. References

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